

◆ BASIC AGGREGATION FUNCTIONS (1–15)

1. SUM

Definition: Adds all numbers in a column

Total Sales = SUM(Sales[Amount])

→ Adds all values from Amount

2. AVERAGE

Avg Sales = AVERAGE(Sales[Amount])

→ Calculates mean value

3. MIN

Min Sale = MIN(Sales[Amount])

→ Smallest value

4. MAX

Max Sale = MAX(Sales[Amount])

→ Largest value

5. COUNT

Count Rows = COUNT(Sales[Amount])

→ Counts numeric rows

6. COUNTA

Count Non Blank = COUNTA(Sales[Customer])

→ Counts non-empty rows

7. COUNTBLANK

Blank Count = COUNTBLANK(Sales[Discount])

→ Counts blank values

8. DISTINCTCOUNT

Unique Customers = DISTINCTCOUNT(Sales[CustomerID])

→ Counts unique values

9. SUMX

Total Revenue = SUMX(Sales, Sales[Qty] * Sales[Price])

→ Row-by-row calculation

10. AVERAGEX

Avg Revenue = AVERAGEX(Sales, Sales[Qty] * Sales[Price])

11. MINX

Min Value = MINX(Sales, Sales[Qty] * Sales[Price])

12. MAXX

Max Value = MAXX(Sales, Sales[Qty] * Sales[Price])

13. DIVIDE

Profit Margin = DIVIDE([Profit], [Sales], 0)

→ Safe division (avoids divide-by-zero)

14. ABS

Absolute Value = ABS(Sales[Profit])

15. ROUND

Rounded Value = ROUND([Avg Sales], 2)

◆ FILTER & CONTEXT FUNCTIONS (16–35)

16. CALCULATE ★ MOST IMPORTANT

Sales 2024 = CALCULATE([Total Sales], YEAR(Sales[Date]) = 2024)

→ Changes filter context

17. FILTER

High Sales = FILTER(Sales, Sales[Amount] > 10000)

18. ALL

Total Sales All = CALCULATE([Total Sales], ALL(Sales))

→ Removes filters

19. ALLEXCEPT

Sales by Region = CALCULATE([Total Sales], ALLEXCEPT(Sales, Sales[Region]))

20. VALUES

Distinct Products = VALUES(Sales[Product])

21. SELECTEDVALUE

Selected Year = SELECTEDVALUE(Date[Year], "Multiple")

22. KEEPFILTERS

Filtered Sales = CALCULATE([Total Sales], KEEPFILTERS(Sales[Amount] > 5000))

23. REMOVEFILTERS

Sales No Filter = CALCULATE([Total Sales], REMOVEFILTERS(Date))

24. ISFILTERED

Is Date Filtered = ISFILTERED(Date[Date])

25. HASONEVALUE

Single Product = HASONEVALUE(Sales[Product])

26. HASONEFILTER

One Filter = HASONEFILTER(Sales[Region])

27. ALLSELECTED

Selected Total = CALCULATE([Total Sales], ALLSELECTED(Sales))

28. CROSSFILTER

Sales One Way = CALCULATE([Total Sales], CROSSFILTER(Sales[ID], Product[ID], ONESIDE))

29. USERELATIONSHIP

Sales Using Alt Date = CALCULATE([Total Sales], USERELATIONSHIP(Sales[ShipDate], Date[Date]))

30. TREATAS

Sales by Year = CALCULATE([Total Sales], TREATAS({2024}, Date[Year]))

31. RELATED

Product Name = RELATED(Product[Name])

32. RELATEDTABLE

Order Count = COUNTROWS(RELATEDTABLE(Orders))

33. EARLIER

Rank Sales = RANKX(Sales, Sales[Amount], , DESC)

34. RANKX

Sales Rank = RANKX(ALL(Sales), [Total Sales])

35. TOPN

Top 5 Products = TOPN(5, Sales, Sales[Amount], DESC)

◆ TIME INTELLIGENCE (36–55)

36. TOTALYTD

YTD Sales = TOTALYTD([Total Sales], Date[Date])

37. TOTALMTD

MTD Sales = TOTALMTD([Total Sales], Date[Date])

38. TOTALQTD

QTD Sales = TOTALQTD([Total Sales], Date[Date])

39. SAMEPERIODLASTYEAR

Last Year Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Date[Date]))

40. DATEADD

Previous Month = CALCULATE([Total Sales], DATEADD(Date[Date], -1, MONTH))

41. DATESYTD

Dates YTD = DATESYTD(Date[Date])

42. DATESMTD

Dates MTD = DATESMTD(Date[Date])

43. DATESQTD

Dates QTD = DATESQTD(Date[Date])

44. PARALLELPERIOD

Sales Last Year = CALCULATE([Total Sales], PARALLELPERIOD(Date[Date], -1, YEAR))

45. PREVIOUSMONTH

Prev Month Sales = CALCULATE([Total Sales], PREVIOUSMONTH(Date[Date]))

46. PREVIOUSYEAR

Prev Year Sales = CALCULATE([Total Sales], PREVIOUSYEAR(Date[Date]))

47. NEXTMONTH

Next Month Sales = CALCULATE([Total Sales], NEXTMONTH(Date[Date]))

48. NEXTYEAR

Next Year Sales = CALCULATE([Total Sales], NEXTYEAR(Date[Date]))

49. STARTOFYEAR

Start Year Date = STARTOFYEAR(Date[Date])

50. ENDOFYEAR

End Year Date = ENDOFYEAR(Date[Date])

◆ LOGICAL FUNCTIONS (51–65)

51. IF

Sales Status = IF([Total Sales] > 100000, "High", "Low")

52. SWITCH

Grade =

SWITCH(TRUE(),

[Sales] > 100000, "A",

[Sales] > 50000, "B",

"C")

53. AND

Result = AND(Sales[Amount] > 5000, Sales[Qty] > 10)

54. OR

Result = OR(Sales[Region] = "North", Sales[Region] = "South")

55. NOT

Not Active = NOT(Sales[Status] = "Active")

56. TRUE

Always True = TRUE()

57. FALSE

Always False = FALSE()

58. ISBLANK

Check Blank = ISBLANK(Sales[Discount])

59. COALESCE

Final Discount = COALESCE(Sales[Discount], 0)

60. IFERROR

Safe Division = IFERROR([Profit]/[Sales], 0)

◆ TEXT FUNCTIONS (66–80)

61. CONCATENATE

Full Name = CONCATENATE(Customer[First], Customer[Last])

62. CONCATENATEX

Product List = CONCATENATEX(Sales, Sales[Product], ", ")

63. LEFT

Left Text = LEFT(Customer[Name], 4)

64. RIGHT

Right Text = RIGHT(Customer[Name], 3)

65. MID

Middle Text = MID(Customer[Name], 2, 4)

66. LEN

Length = LEN(Customer[Name])

67. UPPER

Upper Case = UPPER(Customer[Name])

68. LOWER

Lower Case = LOWER(Customer[Name])

69. TRIM

Clean Text = TRIM(Customer[Name])

70. SUBSTITUTE

Replace Text = SUBSTITUTE(Customer[Name], "-", "")

◆ MATHEMATICAL & STATISTICAL (81–100)**71. POWER**

Square = POWER(4,2)

72. SQRT

Root = SQRT(16)

73. MOD

Remainder = MOD(10,3)

74. INT

Integer = INT(12.7)

75. CEILING

Ceil Value = CEILING(12.3,1)

76. FLOOR

Floor Value = FLOOR(12.9,1)

77. RAND

Random = RAND()

78. RANDBETWEEN

Random No = RANDBETWEEN(1,100)

79. VAR

Variance = VAR.S(Sales[Amount])

80. STDEV

Std Dev = STDEV.S(Sales[Amount])

◆ 81. PATH

Used for: Parent-Child hierarchy (Employee → Manager)

Sample Table: Employees

EmpID	ManagerID
1	
2	1
3	2

DAX

Hierarchy Path = PATH(Employees[EmpID], Employees[ManagerID])

Output

EmpID	Path
1	1
2	1
3	1

◆ 82. PATHITEM

Extracts value from PATH

Level 2 Manager =

PATHITEM([Hierarchy Path], 2)

→ Returns **2nd level manager**

◆ 83. GENERATE

Creates combinations of two tables

Table A

Category
A
B

Table B

Year
2023
2024

Generated Table =

GENERATE(A, B)

Result

Category	Year
A	2023
A	2024
B	2023
B	2024

◆ 84. ADDCOLUMNS

Adds calculated column to a table

Sales With Profit =

ADDCOLUMNS(

Sales,

"Profit", Sales[Revenue] - Sales[Cost]

)

→ Adds Profit column dynamically

◆ 85. SUMMARIZE

Creates summary table

Sales Summary =

SUMMARIZE(

Sales,

Sales[Region],

"Total Sales", SUM(Sales[Amount])

)

Output

Region Total Sales

◆ 86. GROUPBY

Advanced grouping with expressions

Sales Group =

GROUPBY(

Sales,

Sales[Region],

"Max Sale", MAXX(CURRENTGROUP(), Sales[Amount])

)

→ Finds max sale per region

◆ 87. INTERSECT

Common rows between two tables

Common Products =

INTERSECT(OnlineSales, OfflineSales)

→ Returns products sold in both

◆ 88. UNION

Combine tables (no duplicates removed)

All Sales =

UNION(OnlineSales, OfflineSales)

→ Appends rows

◆ 89. EXCEPT

Rows in first table NOT in second

Only Online =

EXCEPT(OnlineSales, OfflineSales)

◆ 90. NATURALINNERJOIN

Join tables automatically on common column

Products

ProductID Name

Sales

ProductID Amount

Joined Table =

NATURALINNERJOIN(Products, Sales)

→ Joins using ProductID

◆ 91. LOOKUPVALUE

VLOOKUP in DAX

Product Name =

LOOKUPVALUE(

Products[Name],

Products[ProductID], Sales[ProductID]

)

→ Fetch product name

◆ 92. DATATABLE

Create table manually

Year Table =

DATATABLE(

"Year", INTEGER,

{

{2023},

{2024}

}

)

◆ 93. FORMAT

Formats value as text

Formatted Sales =

FORMAT([Total Sales], "₹#,##0")

◆ 94. UTCNOW

Returns current UTC time

Current UTC = UTCNOW()

◆ 95. TODAY

Returns today's date

Today Date = TODAY()

◆ 96. NOW

Returns date + time

Current DateTime = NOW()

◆ 97. YEAR

Year Value = YEAR(Sales[Date])

◆ 98. MONTH

Month Value = MONTH(Sales[Date])

◆ 99. DAY

Day Value = DAY(Sales[Date])

◆ 100. WEEKDAY

Returns day number

Day No = WEEKDAY(Sales[Date])

→ Sunday = 1 (default)

◆ 101. EOMONTH

Returns last date of month

End of Month = EOMONTH(Sales[Date], 0)